Four-energy complementarity can alleviate the energy crisis on islands: Case of the Zhoushan Archipelago

Abstract

1 Introduction

As the global economy continues to expand and energy demand rises, fossil fuel–based power generation is increasingly unable to sustain long-term development. This makes the development and utilization of new energy sources a central task in the global energy transition [1],[2]. The oceans—covering about 71% of the Earth's surface—hold enormous renewable energy potential [1],[2], yet major marine resources such as wave and wind energy exhibit significant seasonal and stochastic variability [3]. Their inherent intermittency limits the reliability of single-source marine power systems. To address this issue, the concept of multi-energy complementary systems (MECS) has been proposed [4], and early studies have already demonstrated the advantages of combining different renewable resources, such as wind–solar complementation [5],[6].

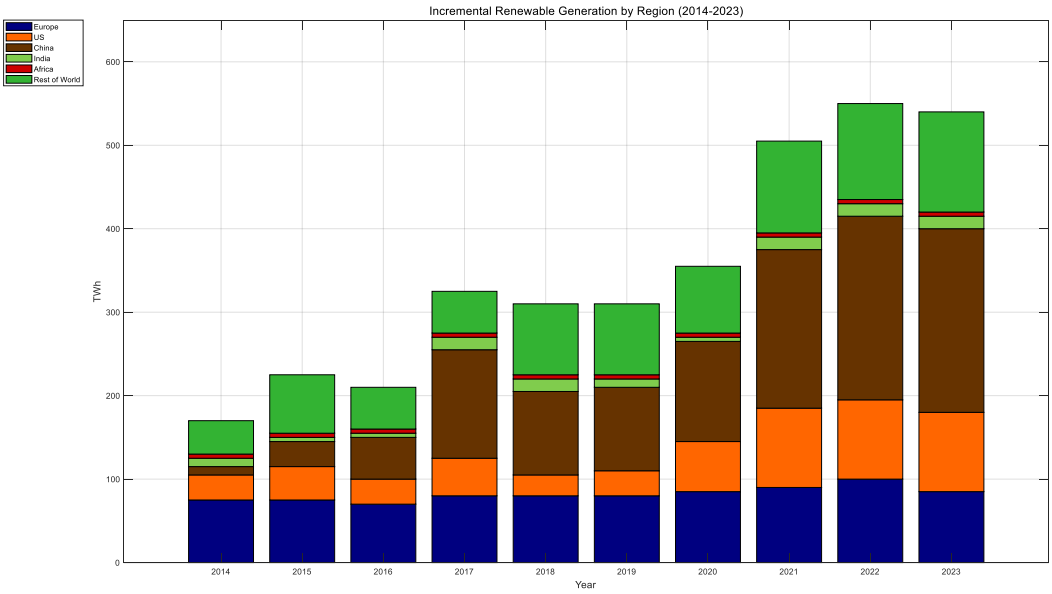


Fig. 1. Renewable Energy Growth Trends (2014-2023)

To address this issue, the study proposes the concept of Multi-Energy

Complementary Systems (MECS). Early research has demonstrated the advantages of combining different renewable resources, such as wind and solar complementarity. For instance, research from the Instituto Tecnológico de Canarias indicates that integrating wind power with hydro storage can significantly increase renewable energy penetration in isolated island systems[7]. This approach optimizes capacity ratios to mitigate output fluctuations, reduce storage requirements, and minimize curtailment[8]. As research advances, additional metrics have been introduced to evaluate the scientific rigor and reliability of offshore multi-energy complementary systems[9][11][13].

Multiple ocean energy pilot projects have been launched globally, but many early attempts have failed due to technical limitations. These failures can be primarily attributed to three categories: technical challenges, commercial constraints, and permitting and regulatory issues (Fig. 2). These experiences provide valuable lessons for the design and evaluation of future MECS demonstration projects, particularly in assessing their long-term operational viability. In such assessments, multiple aspects must be considered, including energy utilization efficiency [12], power generation performance, economic feasibility, and curtailment rates [14].

Fig. 2. Aguçadoura Wave Farm (left) due to technical reasons, Pelamis Wave Energy (center) due to commercial issues, Roosevelt Island Tidal Energy (right) due to permitting and regulatory challenges

Evaluating MECS typically employs the capacity factor (CF) to measure energy utilization efficiency and utilizes the standard deviation (SD) of the composite CF to assess system stability [11,13] [15]. However, traditional evaluation and optimization methods suffer from limitations such as: focusing solely on CF or SD may compromise average power output; conventional weighting methods involve subjectivity; and heuristic algorithms require trade-offs between computational complexity and solution accuracy [17,18]. To address these challenges, this study introduces the Entropy Weighting Method (EWM), an effective tool for assigning objective weights based on the information content and variability of metrics [19–21]. Integrating EWM into MECS evaluation minimizes the subjectivity of traditional weighting, achieving an

objective and adaptive balance between utilization efficiency (high CF) and system stability (low SD) [20].

Accordingly, this study applies EWM to quantify the proportional contribution of wind, wave, tidal, and solar energy within a marine MECS in the Zhoushan Archipelago, China. By weighting each energy source according to the temporal variability of its capacity factor, the method achieves an optimal trade-off between utilization efficiency and system stability. Sensitivity analyses, including curtailed energy and capture price, are further conducted to evaluate economic feasibility and the potential for mitigating price self-cannibalization.

2 Materials and Methods

This chapter aims to elaborate in detail on the resource data employed in this study, the characteristics of the research area, and the core methodology for optimizing the configuration of multi-energy complementary systems. The scientific rigor and reliability of the research are grounded in precise data inputs and a rigorous modeling framework, ensuring the scientific validity of the configuration results.

2.1 Study area

The study area covers the waters surrounding the Zhoushan Archipelago, extending from latitudes 29.4°–31.0°N and longitudes 121.5°–123.5°E, as shown in Fig. 3. Located in the East China Sea, the Zhoushan Archipelago is the largest island group in China, comprising 1,390 islands — the land area accounts for approximately 20% of China's total island territory — and hosting a population of approximately one million [22]. The region features a complex coastal and archipelagic geomorphology, characterized by strong and persistent wind resources [23]. In addition to being influenced by the northwest Pacific winds, it also possesses enormous wave energy potential from swell activity [24]. Moreover, the tidal energy resources in the Zhoushan Archipelago are exceptionally significant, with a theoretical average power capacity reaching 5.6×10^6 kW, accounting for over half of China's total tidal energy resources. [25]. Annual sunshine duration surpasses 2,000 hours, providing abundant solar radiation. This exceptional combination of natural endowments creates highly favorable conditions for the integrated development of wind, wave, tidal, and solar energy, making the area a promising hub for offshore renewable energy utilization.

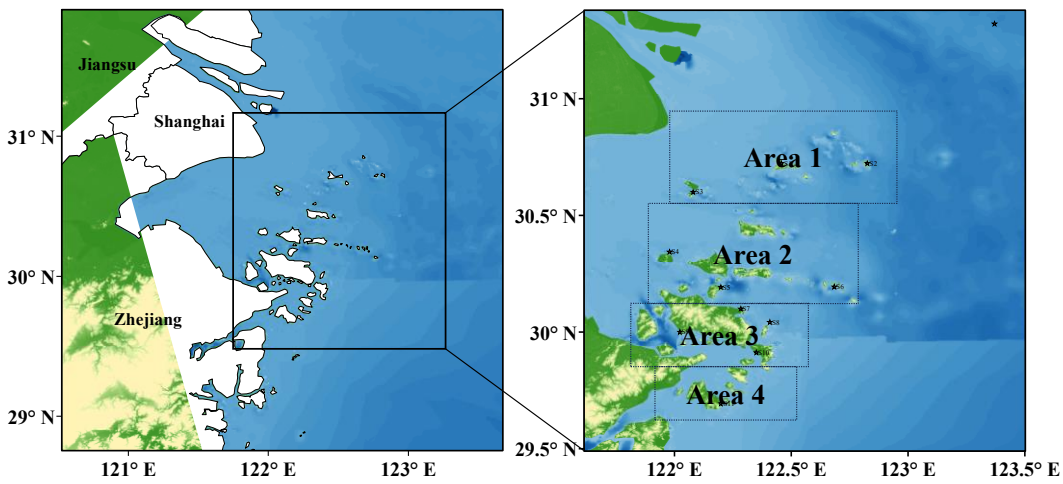


Fig. 3. Location of the study area and the high-potential development zones delineated under the pilot project.

In recent years, the Chinese government has launched a series of multi-energy complementary pilot projects in the Zhoushan Archipelago to explore the coordinated development and efficient utilization of wind energy, solar energy, tidal energy, and natural gas resources, among others. Representative projects include the megawatt-scale tidal energy demonstration project (S5) on Xiushan Island and the self-healing smart island microgrid project (S6) on Dongji Island, both of which successfully integrated tidal energy with photovoltaic power generation. These initiatives have provided continuous, stable power supply to island communities, alleviating local electricity shortages and enhancing regional energy security. Most of these pilot projects are still undergoing technological upgrades, serving as long-term experimental platforms for offshore energy integration and equipment testing.

Building upon the achievements and experiences of these pioneering projects, this study systematically identifies potential areas for multi-energy complementary development within the Zhoushan Archipelago. Although few previous studies have proposed specific spatial zoning frameworks for this purpose, this research establishes a scientific four-zone plan based on a comprehensive analysis of existing project distributions, spatial variations in energy resource endowments, and regional grid structures. Fig. 3 displays the locations of existing projects and the four designated development areas, providing a rational and practice-oriented spatial framework for optimizing future offshore multi-energy complementary systems.

To systematically guide and advance regional deployment of MECS, this study divides the Zhoushan Archipelago into four high-potential development areas (Area 1–4) based on prior pilot projects. The zoning framework primarily follows administrative

boundaries and geographic clustering, providing a clear spatial foundation for subsequent system optimization.

This delineated zoning enables focused analysis of regional variations in energy resource availability, infrastructure conditions, and energy demand across four areas. This facilitates more efficient and cost-effective optimization of MECS and allows for the formulation of tailored development strategies for different regions based on the findings.

2.2 Data

The foundational resource assessment utilizes the ERA5 reanalysis dataset from the European Centre for Medium-Range Weather Forecasts (ECMWF), supplemented by measured wave data from the East China Sea Wave Observation Station, tidal level and current velocity data from the Zhoushan Daishan Tide Gauge Station, and topographic datasets from NOAA and GEBCO [26]. Simulation data were generated using the SWAN and MIKE numerical models and validated against field observations. Additionally, technical specifications were collected from domestic and international manufacturers of wind turbines, wave energy converters, tidal energy devices, and photovoltaic modules [27][28]. All datasets underwent model validation and standardization in accordance with national and industry standards to ensure scientific reliability and consistency.

2.3 Analytical Methods

The capacity factor (CF), defined as the ratio between the actual power output of a device and its rated maximum output over a given period[29], serves as a fundamental indicator for characterizing the utilization efficiency and temporal variability of renewable energy resources. Unlike instantaneous power or resource availability alone, CF inherently reflects both resource conditions and conversion performance, making the CF time series an ideal basis for evaluating the dynamic behavior of multi-energy systems. The formula for entropy rights calculation is provided in Appendix A.

Accurately determining the relative importance of different energy sources is essential for evaluating the overall performance of a multi-energy complementary system. Traditional weighting approaches often rely on subjective judgment or fixed assumptions, making them insufficient for characterizing the strong temporal variability inherent in renewable resources. To obtain an objective, data-driven set of weights, this study applies an EWM based on the information content of the hourly capacity factor (CF) time series.

Let $X = \{x_{ij}\}_{m\times n}$denote the original CF matrix, where x_{ij}represents the CF of the j-th energy type at hour i. After applying min–max normalization and introducing a small correction constant $\varepsilon = 10^{-6}$ to avoid logarithmic singularities, the standardized matrix $R = \{r_{ij}\}$is obtained. The proportion coefficient for each indicator is then defined as

$$p_{ij} = \frac{r_{ij}}{\sum_i r_{ij}} \tag{1}$$

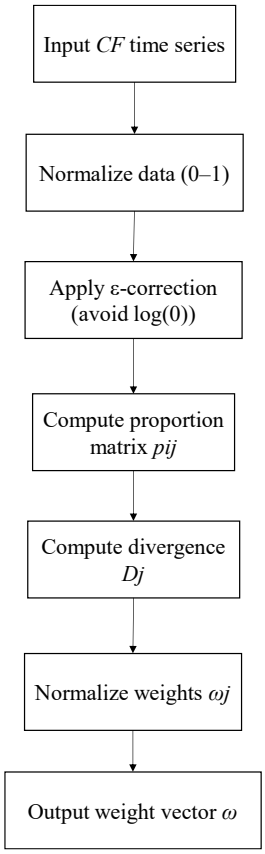
Based on information theory, the entropy value of the j-th indicator is computed as

$$E_j = -k\sum_{i=1}^{m} p_{ij} \ln p_{ij} \tag{2}$$

where $k = 1/\ln m$. A smaller entropy value reflects larger temporal variability and thus a greater information contribution. The divergence coefficient is defined as $D_j = 1 - E_j$, and the final normalized weight is obtained as

$$\omega_j = \frac{D_j}{\sum_j D_j} \tag{3}$$

The complete procedure, including data preprocessing, entropy calculation, and weight normalization, is illustrated in Fig. 4, accompanied by pseudocode for reproducibility. Compared to traditional subjective weighting or single-indicator optimization, the enhanced entropy method provides a stable, fully objective weighting mechanism that directly responds to the intrinsic variability of each energy source. This ensures that highly fluctuating resources—those contributing more information to the overall system behavior—receive proportionally higher weights, leading to a more balanced and scientifically robust evaluation framework.

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Algorithm 1: Enhanced Entropy Weight Method

Input: CF time-series X(m×n)
Output: Weight vector ωj

Normalize X to obtain R
Add ε = 1e-6 to avoid log(0)
for each indicator j do
    Compute pij = rij / sumi(rij)
    Compute entropy Ej = -k * Σ(pij ln pij)
    Compute Dj = 1 - Ej
end for
Compute ωj = Dj / Σ Dj
return ω
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Fig. 4. Entropy-Based Rights Calculation Process

In addition to determining source weights, the CF time series and the resulting weighted composite output also enable further analyses of system behavior under operational constraints. These include curtailment patterns, economic performance, and capture-price variability. While the detailed sensitivity analysis is presented later, the entropy-based weighting established here forms the analytical basis for assessing how different resource combinations influence both technical stability and economic outcomes.

2.4 Complementary analysis

Determining the optimal energy allocation ratio in MECS is a critical step toward improving overall stability and utilization efficiency. However, traditional optimization strategies exhibit several inherent drawbacks. They frequently depend on historical data to establish fixed weighting schemes, which limits adaptability to dynamic fluctuations in renewable energy resources—such as seasonal variations in wind or solar capacity factors. Moreover, subjective or empirical weight assignments often lack a rigorous quantitative foundation, reducing the scientific credibility of the resulting allocation results. In addition, the lack of flexible threshold adaptation mechanisms often leads to computational inefficiency and convergence difficulties in optimization procedures.

To address these limitations, this study integrates the enhanced entropy weight method with a complementary optimization framework. The objective is to identify an allocation scheme that achieves a balance between system stability and energy utilization efficiency.

Within conventional optimization paradigms, the Minimum Standard Deviation (MSD) approach is commonly employed, in which system stability is maximized by minimizing fluctuations in the total capacity factor (CF_t). The CF_t of the MECS at time t is defined as:

$$CF_t = \sum_{i=1}^{n} \omega_i \cdot CF_{i,t} \tag{4}$$

where $CF_{i,t}$ represents the capacity factor of the i-th energy source at time t, and ω_i is the corresponding entropy weight derived from Section 2.3.

Furthermore, the system's stability is evaluated through the SD of the CF_t, given by:

$$\sigma = \sqrt{\frac{1}{N}\sum_{t=1}^{N}(CF_t - \overline{CF_t})^2} \tag{5}$$

Where N is the number of time intervals, CF_t represents the total capacity factor at time t, and $\overline{CF_t}$ is its mean value. A smaller σ indicates higher temporal stability and stronger complementarity among energy sources.

By combining the improved entropy weighting approach with the complementary performance evaluation framework, this study provides an adaptive and objective methodology for energy allocation. The model enables dynamic response to temporal variability in renewable energy resources, ensuring both high utilization efficiency and stable system output—achieving a balanced optimization objective unattainable by conventional single-criterion methods. The formulas for sensitivity analysis calculations are provided in Appendix B.

3 Complementarity Potential Analysis

This study analyzes the annual variation patterns and complementary characteristics of multiple renewable energy sources within the Zhoushan Archipelago area based on the total capacity factor (CF_t). CF_t and its SD are employed to measure energy utilization efficiency and output variability, with the former reflecting installed capacity utilization levels and the latter indicating system stability. An ideal multi-energy system should maintain a high CF_t while keeping SD low, thereby achieving

stable and efficient energy supply.

3.1 Temporal Characteristics of Capacity Factors

The annual variations of the four energy sources, as shown in Fig. 5, display significant seasonal differences. Wind energy peaks during winter (December to February) and declines markedly in summer. Wave energy follows a highly consistent temporal pattern with wind energy, increasing in winter alongside enhanced wind conditions. Solar energy reaches its maximum during summer (June to August), with irradiation significantly weakening in winter, demonstrating a typical inverse relationship with wind energy. Tidal energy, governed by astronomical cycles, maintains relatively stable annual output with minimal seasonal fluctuations. The quarterly box plot (Fig. 6) further indicates that wind energy exhibits the highest variability, with wave energy containing a significant number of outliers due to typhoon impacts. Solar shows some fluctuation and contains partial outliers, while tidal energy demonstrates the lowest dispersion and highest stability. Overall, the Zhoushan sea area presents a temporal pattern characterized by "simultaneous peaks in wind and wave energy, coupled with an inverse relationship in photovoltaic output," providing a natural resource foundation for achieving multi-energy complementarity.

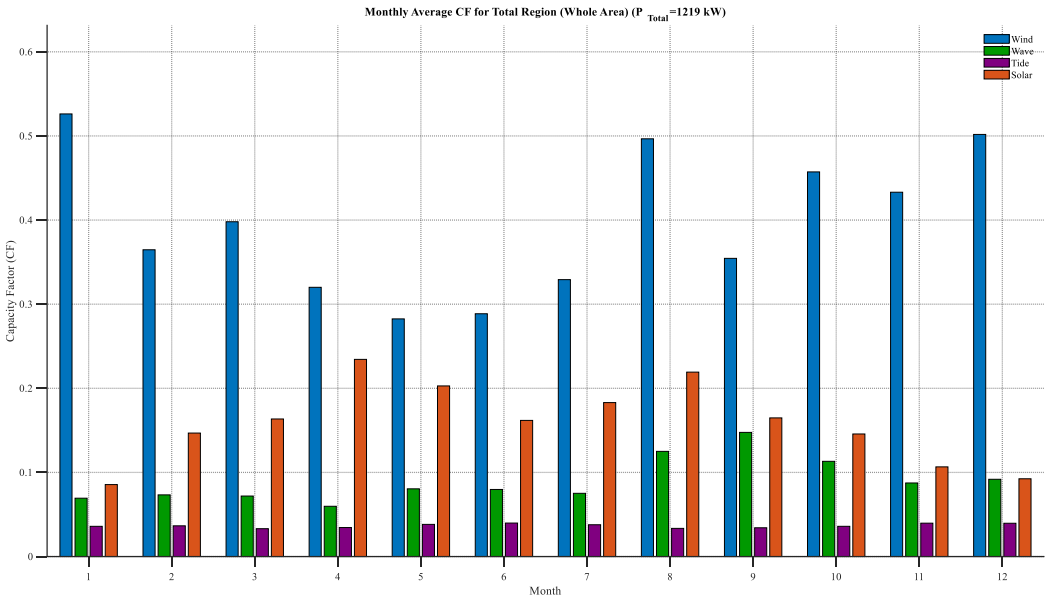

Fig. 5. Monthly CF Change Curves for Each Energy Source.

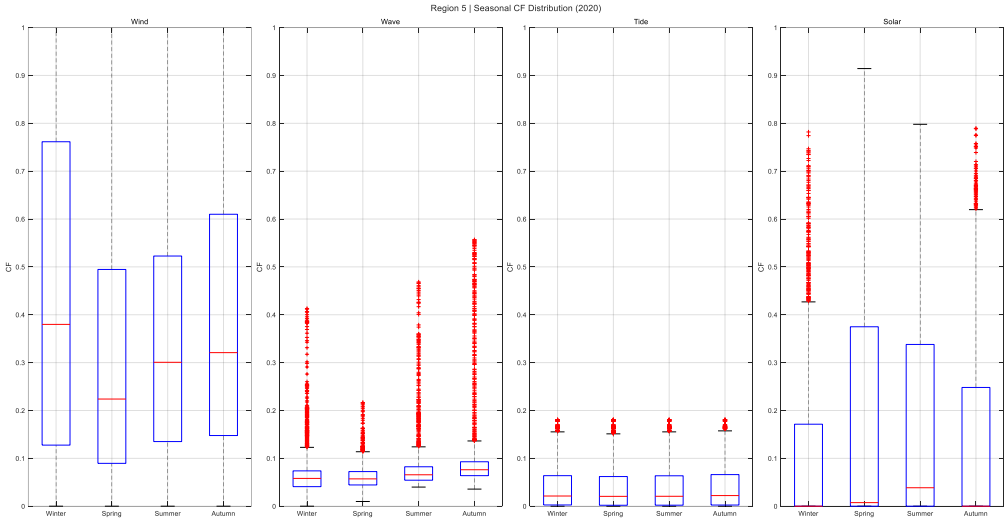

Fig. 6. Quarterly CF Box-and-Line Charts for Each Energy Source.

3.2 Temporal Complementarity Among Energy Sources

The temporal complementarity between different energy sources is quantified using correlation coefficients (Fig. 7). The correlation between wind and wave energy remains consistently high, mostly at 0.5 or above. This is primarily due to higher wind speeds in late winter and early spring, and lower speeds in late spring and early summer. From July to October, frequent typhoon activity in the northwest Pacific generates substantial swells. These swells, combined with locally induced wind waves from coastal storms and large-scale swells from offshore storms, significantly alter wave conditions in the Zhoushan Sea area. Tidal energy production is primarily influenced by celestial gravitational forces and is almost entirely independent of meteorological conditions (wind, waves, solar radiation). Consequently, harnessing tidal energy can effectively mitigate the risks associated with energy instability. This holds significant importance for combined power generation, as it enhances system stability.

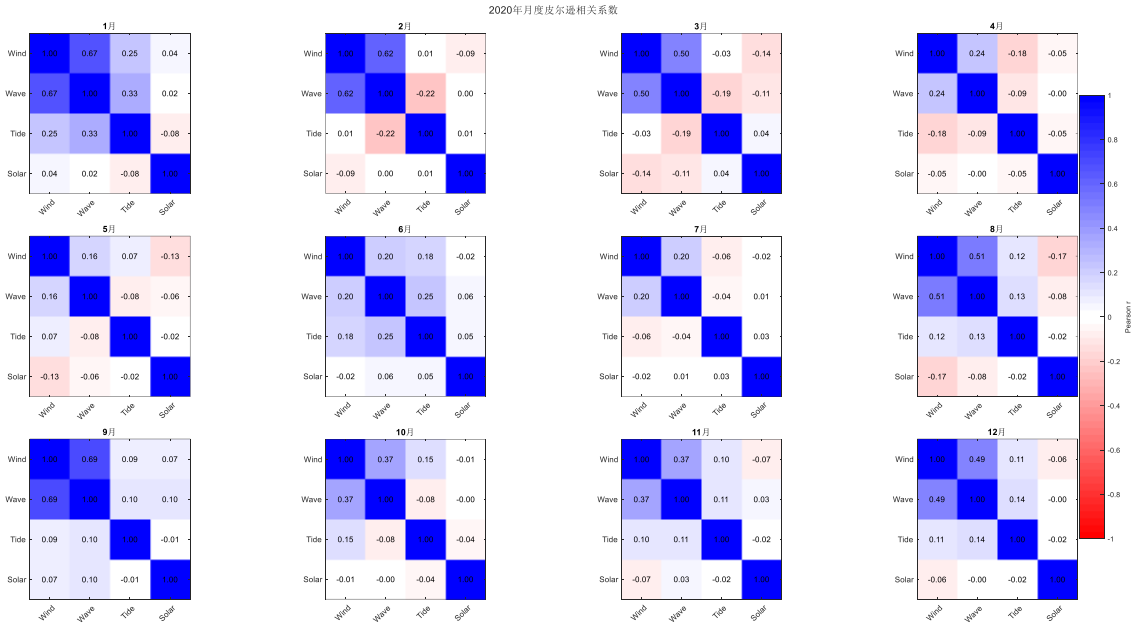

Fig. 7. Monthly Pearson Correlation Coefficients for Various Energy Sources.

4 Results

4.1 Analysis of Computational Results

Based on the indicator weights determined by the entropy weight method, the energy contribution and comprehensive performance of each sub-area were further calculated (**Table 1**). Results indicate that the Zhoushan Archipelago is dominated by tidal energy (31.59%) and solar energy (44.68%), with a system-wide comprehensive CF_t of 0.1495 and a SD of 0.1114. Significant variations were observed across sub-regions: Area 1 exhibits substantial wave energy influence (62.64%), yet demonstrates poor CF_t and steady-state performance (0.0373 and 0.0357, respectively); Areas 2 and 3 rely primarily on tidal and solar energy, with CF_t values consistently ranging between 0.145 and 0.153; Area 4 exhibits a higher proportion of wave energy (34.93%) and demonstrates overall performance in the middle range.

Table 1

Calculated CF_t and σ of the MECS based on the EWM.

Area	Wind	Wave	Tide	Solar	CF_t	σ
Total	14.84%	8.90%	31.59%	44.68%	0.1495	0.1114
Area 1	9.10%	62.64%	14.56%	13.70%	0.0373	0.0357
Area 2	16.06%	8.82%	34.44%	40.68%	0.1454	0.1085
Area 3	15.90%	11.04%	33.18%	39.88%	0.1526	0.1097
Area 4	12.00%	34.93%	24.59%	28.49%	0.1188	0.0850

Further analysis combining power output and economic indicators (**Table 2**)

reveals that Area 4 exhibits the highest average power and total generation capacity, yet also faces higher curtailment risks and lower capture prices. This indicates that resource development in this area requires supporting energy storage or flexible loads (such as hydrogen production and desalination) to absorb fluctuating output [44]. Area 2 and Area 3 achieve a favorable balance between stability and generation efficiency, making them suitable as priority areas for grid-connected development. While Area 1 exhibits the lowest variability, its output remains relatively low, rendering it unsuitable as a primary development zone.

Table 2

Power generation efficiency, electricity generation volume, and average power output by region.

Area	P_{Rated}(kW)	E_{Total}(MWh)	P_{Avg}(kW)
Total	1218.7	1600.3	182.19
Area 1	4105.1	1345.1	153.13
Area 2	1271	1623.3	184.8
Area 3	1391.8	1865.4	212.37
Area 4	2617.6	2732.4	311.07

To compare the advantages and disadvantages between the entropy weight method and traditional approaches, we introduce energy weights calculated using the MSD method. In contrast, while the MSD method (**Table 3**) significantly reduces volatility, it tends to result in extreme weight distributions and decreased system utilization. The entropy weight method, however, achieves a balance between stability and efficiency by measuring differences in information entropy metrics, demonstrating more reasonable engineering applicability.

Table 3

Calculated CF_t and σ of the MECS based on the MSD method.

Area	Wind	Wave	Tide	Solar	CF_t	σ
Area 1	0.00171190%	99.90%	0.07%	0.03%	0.0006	0.0006
Area 2	0.00000083%	79.96%	18.13%	1.91%	0.0479	0.0327
Area 3	0.00000002%	84.20%	13.17%	2.63%	0.0464	0.0370
Area 4	0.00000603%	97.71%	1.31%	0.98%	0.0181	0.0177

The MSD method, in its pursuit of lowest volatility, exhibits an extremely concentrated weighting distribution. For instance, in Area 1, nearly all weight is allocated to wave energy, while wind, solar, and tidal energy account for negligible shares. Similarly, Area 4's mix is overly reliant on wave energy (97.71%). In contrast,

the entropy-weighted method yields a more balanced allocation, particularly in rows for Area 2 and Area 3, where wave energy serves as the primary source. This aligns with the natural complementary characteristics of the Zhoushan Archipelago's overall resource endowment.

Calculations of power generation and other metrics based on the MSD method are shown in **Table 4**. Although the instantaneous power P (kW) in each area remains at a relatively high level—for example, peak power in Area 1 and Area 4 reached approximately 5,894.4 kW and 5,768.9 kW respectively, demonstrating strong short-term energy density— However, further analysis combining annual energy output E(MWh) and average power PAvg(kW) reveals that the actual available capacity in these areas is significantly weaker. Taking Area 1 as an example, despite a peak approaching 6 MW, its annual energy output is only 8.426 MWh, corresponding to an average power of less than 1 kW. Other areas similarly exhibit the characteristic of "strong peaks but weak averages." This indicates that these areas exhibit high energy fluctuations and low stability, making them unsuitable for sustained power generation or long-term deployment. Therefore, employing the entropy weight method for resource allocation analysis can further reduce volatility.

Table 4

Calculated CF_t and σ of the MECS based on the MSD.

Area	P(kW)	E(MWh)	P_Avg(kW)
Area 1	5894.4	8.426	0.95924
Area 2	4772.3	2392.6	272.38
Area 3	5007.6	2236.6	254.62
Area 4	5768.9	605.15	68.892

Regarding the trade-off between stability and utilization rate, the MSD method achieves the lowest std in Area 1 and Area 4, significantly lower than the entropy-weighted method. This reflects its absolute optimization for stability. However, while achieving low std, the MSD method often leads to a significant reduction in CF_t. For example, Area 1's CF_t is only 0.0006, far below the entropy-weighted method's optimized result. This confirms our earlier judgment: solely targeting minimum std may sacrifice the system's average output and energy utilization efficiency due to overly concentrated allocation.

Furthermore, **Table 1** shows that the Zhoushan Archipelago primarily targets solar and tidal energy as development sources, with wind and wave energy accounting for

relatively lower shares. Although solar energy exhibits the lowest capacity factor in winter, its inherent seasonal inverse correlation with wind energy's variability confirms solar's potential as a key component in multi-energy complementary systems—providing robust support for pilot project development. Concurrently, tidal energy's consistently high CF levels throughout the year and low variability yield advantages of low information entropy and high weighting, making it crucial for system stability. Wave energy's CF follows a seasonal pattern similar to wind energy—higher in autumn and winter, lower in summer—with significant variability, thus receiving the lowest weighting.

Based on the objective weights derived from the entropy weighting method, a configuration scheme can be established with solar and tidal energy as the main sources and wind and wave energy as supplementary sources, aiming to achieve an optimized balance between utilization rate and stability. This objective weighting method effectively avoids the subjectivity issues in weight determination inherent in traditional configuration methods.

Evaluation results for typical pilot sites (S1–S4) (**Table 5**) further validate regional differences. CF_t values at each site ranged from 0.09 to 0.11, with *SD* below the regional average. Site S1 and S4 benefited significantly from tidal energy contributions, S3 showed dominance of wave energy, and S2 featured a higher proportion of solar power. Overall, the site-level analysis aligns with the pilot experiment's direction, providing robust scientific support for the previous pilot project.

Table 5

Calculation results of typical pilot projects based on the EWM.

Area	Wind	Wave	Tide	Solar	CF_t	σ
S1	13.20%	0.00%	56.70%	30.11%	0.1035	0.0821
S2	12.06%	13.17%	44.70%	30.07%	0.1115	0.0837
S3	9.25%	49.37%	19.66%	21.72%	0.0882	0.0636
S4	9.81%	4.95%	58.03%	27.21%	0.0997	0.0724

4.2 Sensitivity Analysis

To further assess economic impacts, a sensitivity analysis was conducted on curtailed electricity volume and captured electricity prices (**Table 6**). Results indicate that Area 4 exhibits the highest curtailment rate (21.52%) and lowest captured electricity price (0.3186 yuan/kWh), reflecting market discounts caused by supply surplus. Area 2 demonstrates the lowest curtailment rate (6.08%) and highest captured electricity price (0.3733 yuan/kWh), highlighting its excellent load matching

characteristics. After entropy weight optimization, the overall price discount rate for the system decreased from 20.35% to 17.77%. and the highest captured electricity price (0.3733 yuan/kWh), demonstrating its excellent load matching characteristics. After optimization using the entropy weight method, the overall price discount rate of the system decreased from 20.35% to 6.67%, significantly improving economic returns. This indicates that multi-energy complementary configurations not only enhance system operational stability but also effectively mitigate the price erosion effect of renewable energy in the market.

Table 6

Wasted electricity volume, wasted electricity rate, actual annual total power generation, capture cost.

Area	$E_{Curtailed}$ (MWh)	Curtailment Rate	E_{Sold} (MWh)	Pcap(Yuan/kWh)
Total	96.88	6.05%	1503.5	0.3729
Area 1	95.15	7.07%	1250.0	0.3693
Area 2	98.62	6.08%	1524.6	0.3733
Area 3	161.90	8.68%	1703.5	0.3650
Area 4	587.93	21.52%	2144.5	0.3186

In summary, the landscape resources of the Zhoushan Archipelago exhibit a typical negative correlation and complementary characteristics on a seasonal scale. The stable output of tidal energy and the winter enhancement of wave energy jointly reinforce the overall synergy of the system. The entropy weight method effectively balances utilization rate and stability in weight allocation, demonstrating greater robustness and practical value in engineering applications compared to the MSD method.

5 Discussion

This study reveals the temporal and spatial complementary characteristics of multiple renewable energy sources—wind, wave, tidal, and solar—in the Zhoushan Archipelago region. Results indicate a significant negative correlation between wind and solar energy on seasonal scales: wind power output peaks in winter, while solar irradiance is strongest in summer. This “winter wind, summer sun” complementary pattern reflects the typical distribution characteristics of monsoon-type ocean energy, providing a natural foundation for constructing regional multi-energy complementary systems.

Fig. 5 displays that pronounced seasonal complementarity primarily stems from the annual variation patterns of the East Asian monsoon system. During winter, under the influence of the strong northwest monsoon and land-sea temperature differences,

wind speeds generally increase across the Zhoushan sea area, significantly boosting wind energy output. Conversely, in summer, under the control of the subtropical high-pressure system, reduced cloud cover and improved radiation conditions lead to peak solar energy resources. The inverse correlation between these resources on seasonal scales reflects climate-driven energy conversion patterns. This cyclical variation creates a natural time-shifted relationship between wind and solar resources, effectively mitigating seasonal fluctuations in system output.

Compared with similar domestic and international studies, the findings of this research demonstrate high consistency. Lv et al. (2022) analyzed the spatiotemporal distribution and complementarity of wind and solar energy in China, identifying regions with strong complementarity primarily in the coastal areas of Zhejiang Province within the southeastern river basins [45]. Wen et al. (2025) also demonstrated a seasonally negatively correlated complementary pattern across East Asian coastal areas in their global offshore wind-solar complementarity study [46]. Our results align with these studies but reveal a more pronounced complementarity, potentially attributed to the complex island topography of the Zhoushan Archipelago and the synergistic effects of multiple energy resources. Furthermore, compared to temperate seas like the North Sea and Mediterranean, the East China Sea exhibits stronger monsoon cycles and greater seasonal amplitude in wind energy, further amplifying the negative correlation between wind and solar. This indicates that both climatic background and regional dynamic processes jointly determine the spatiotemporal variations in multi-energy complementarity.

At the multi-energy system level, wave and tidal energy partially compensate for seasonal gaps in wind and solar output. Findings indicate that wave energy increases in tandem with wind energy during winter, while tidal energy exhibits relatively stable seasonal patterns, playing a peak-shaving and valley-filling role during spring and autumn transitions. This multi-energy coupling not only enhances system output stability but also demonstrates the synergistic potential of ocean energy within composite systems. Particularly during months with low wind-solar complementarity, tidal energy's stable output provides foundational load support, delivering significant system value.

Areas 2 & 3, with their exceptional inherent complementarity, are ideal candidates for grid-scale large-scale integration projects. In terms of configuration, maintaining a stable ratio between tidal energy and wind/solar power should be prioritized, allowing

for reduced reliance on external energy storage systems and thereby lowering initial investment costs. Area 4, while possessing immense energy potential, faces high risks of curtailed electricity. Development in this area should adopt technical coupling strategies, such as integrating large-scale energy storage systems. High-capacity battery storage systems can absorb excess peak electricity and shift it for sale during high-price periods. Energy conversion can utilize curtailed electricity for water electrolysis to produce hydrogen or for seawater desalination, enabling cross-sector energy utilization that transforms low-cost electricity into high-value fuels or resources.

From an engineering and planning perspective, this complementarity holds practical significance for island-based energy systems. The wind-solar complementary characteristics in the Zhoushan region can significantly reduce the capacity requirements of energy storage systems: wind power dominates generation in winter, while solar power supplements output in summer, achieving a natural match for seasonal loads. Simultaneously, the introduction of wave and tidal energy further enhances system reliability and utilization rates. This provides a theoretical basis for promoting multi-energy coordinated supply models in island regions in the future and offers scientific reference for formulating regional energy layout plans at the policy level. The core advantage of this method lies in its data-driven and objectively weighted nature, making it applicable to any energy system facing multi-source allocation and volatility challenges. For instance, Gasa Island in South Korea or remote fishing islands in Europe, which similarly rely on multiple marine energy sources, exhibit vastly different temporal characteristics across these resources. By inputting local resource capacity coefficient data, the model can rapidly generate objective and efficient initial allocation schemes.

It should be noted that this study's energy complementarity assessment relies on ERA5 reanalysis data, whose spatial and temporal resolutions may influence results. Additionally, the model does not yet account for the combined effects of energy storage, transmission constraints, and economic indicators, potentially introducing biases in system benefits. Future research will integrate high-resolution observational data with multi-scenario optimization models to further explore the dynamic operational characteristics and economic viability of multi-energy systems.

6 Conclusion

This study established a comprehensive evaluation system for the complementarity of wind, wave, tidal, and solar energy in the Zhoushan Archipelago

region based on ERA5 reanalysis data. Using the entropy weighting method to determine the weights of each complementarity indicator, it quantified the complementary characteristics of multiple marine energy sources across temporal and spatial scales, providing methodological and theoretical support for the coordinated development of offshore renewable energy systems. Key findings are as follows:

Wind and solar energy exhibit a significant negative correlation at seasonal scales in the Zhoushan Archipelago region: wind energy peaks in winter, while solar energy is strongest in summer, demonstrating a typical monsoon-type complementary pattern characterized by “winter wind and summer sun.” This temporal complementarity significantly reduces seasonal fluctuations in system output.

Wave and tidal energy play crucial balancing roles within the multi-energy system. Wave energy increases in winter alongside enhanced wind fields, while tidal energy remains relatively stable, providing peak shaving and valley filling effects during transitional seasons. The synergistic interaction of these four energy sources significantly enhances the system's overall stability, validating the feasibility and advantages of the offshore multi-energy complementary model.

Entropy weighting analysis indicates that system volatility metrics carry the highest weight, confirming stability as the critical performance factor in multi-energy systems. Integrated complementarity index results further reveal that waters surrounding northern and central Zhoushan islands exhibit the highest multi-energy complementarity potential, making them ideal priority zones for offshore integrated energy systems.

These findings provide valuable reference for engineering planning of island-based energy systems. By rationally configuring wind and solar capacity ratios and integrating wave and tidal energy, system reliability and renewable energy utilization can be enhanced without significantly increasing energy storage capacity. This offers scientific basis for advancing the “multi-energy complementary, source-grid-storage coordinated” energy transition on eastern coastal islands in China.

It should be noted that this study did not account for practical operational conditions such as energy storage scheduling, economic constraints, or grid integration. The results primarily reflect the complementary potential at the natural resource level. Future work will incorporate energy storage system simulations, cost-benefit analyses, and multi-scenario optimization models to refine the full-lifecycle assessment framework for offshore multi-energy systems, thereby expanding research from

resource complementarity to operational optimization.

7 Acknowledgements

Appendix A. Calculation Methods for Capacity Factors

Below are the calculation models, parameter settings, and derivation processes for the capacity factors of four energy sources.

A.1. Wind power capacity factor

The wind energy capacity factor can be calculated by following the approach of Wen et al. [30]: first, average the instantaneous power Pt over a period of time (e.g., one year), then divide this average by the theoretical annual energy production of the wind turbine operating continuously at its rated power Pe to obtain the capacity factor.

$$P_t = \begin{cases} 0 & V < Cut\text{-}in\ speed \\ \frac{1}{2}\rho \cdot C_P \cdot A_T \cdot V^3 & Cut\text{-}in\ speed \le V < Rated\ speed \\ P_e & Rated\ speed \le V \le Cut\text{-}out\ speed \\ 0 & V > Cut\text{-}out\ speed \end{cases} \tag{6}$$

$$CF_{wind} = \frac{P_t}{P_e \times 8760} \tag{7}$$

In equation (2.2), ρ represents the air density, taken as 1.225 kg/m³ in this study; Cp denotes the wind energy utilization coefficient, typically set at 0.4 for modern wind turbines; AT is the swept area of the turbine blades, with a blade diameter of 154 m used in this study; V represents the actual wind speed at the turbine hub height, set at 100 m in this study.

A.2. Wave power capacity factor

The wave energy capacity coefficient can be calculated following the approach by Bahareh Kamranzad et al. [31], using the ratio of the actual annual/monthly power generation of the WEC at a specific site to its theoretical maximum energy output, as shown in Equation (2.4):

$$CF_{wave} = \frac{\sum_{i=1}^{n_T}\sum_{j=1}^{n_H} p_{ij}P_{ij}}{P_e \times 8760} \tag{8}$$

A.3. Tidal power capacity factor

As in Equation (2.5) [32], referencing the fan power curve model, replace “wind speed” with “current flow velocity (v)” and “air density” with “seawater density (ρwater = 1025 kg/m³)”.

$$P_t = \begin{cases} 0 & V < Cut\text{-}in\ velocity \\ \frac{1}{2}\rho_{water} \cdot C_P \cdot A_T \cdot V^3 & Cut\text{-}in\ velocity \le V < Rated\ velocity \\ P_e & Rated\ velocity \le V \le Cut\text{-}out\ velocity \\ 0 & V > Cut\text{-}out\ velocity \end{cases} \tag{9}$$

$$CF_{tidal} = \frac{P_t}{P_e \times 8760} \tag{10}$$

In the formula, AT represents the swept area of the tidal turbine; Ptidal denotes the rated power of the tidal energy converter; Cp is the power coefficient of the turbine (derived from rated parameters, following the same logic as wind energy).

A.4. Solar photovoltaic power capacity factor

Solar CF requires integrating the conversion efficiency of photovoltaic panels with actual solar radiation intensity (accounting for temperature and wind speed effects on panel performance). First, calculate the actual average power generation capacity, then compare it with the rated power of the photovoltaic panels. Introduce the "performance ratio PR" to correct for environmental factors. As shown in Equation (2.7) [33].

$$CF_{solar} = P_R \frac{RSDS}{RSDS_{STC}} \tag{11}$$

In equation (2.7), RSDS denotes solar radiation density, and STC represents standard test conditions (RSDSSTC = 1000 W/m²).

Appendix B. Calculation of Curtailment Volume and Capture Costs

Wasted electricity refers to the portion of electricity generated by the system that remains unused due to grid constraints, insufficient transmission capacity, or instantaneous output exceeding local load demand. Its calculation formula is:

$$E_{cur} = \sum_{t=1}^{8784} (\max(0, P_{gen,t} - P_{load,t}) \times \Delta t \tag{12}$$

$P_{gen,t}$ denotes the total power generation of the system at time t, calculated using the formula:

$$P_{gen,t} = \sum_i (CF_{i,t} \times P_{Rated,i}) \tag{13}$$

$P_{load,t}$ denotes the actual load demand or grid capacity at time t, serving as the key constraint determining whether curtailment occurs. Based on Zhang et al. (2018)[38], given that the population density of Beiji Island in China is comparable to that of the

Zhoushan Archipelago, we adopt a peak load of 800 kW as the reference value. Studies by Husein et al. (2017) [39] and Bertheau et al. (2020) [40] indicate that for small islands dominated by low-energy industries (fisheries, agriculture, etc.), the average load is approximately 0.5 times the peak load. Thus, the average load for the Zhoushan Archipelago can be estimated at 400 kW. Additionally, we have adopted a dynamic load calculation method that incorporates morning and evening peak electricity consumption periods. Therefore, a sine function was selected to generate the dynamic load as an idealized representation; Δt denotes the time step, typically set to 1 hour.

The capture price represents the actual weighted average selling price of renewable energy generation. If P_{cap} exceeds the average market electricity price, it indicates that the system's output aligns well with high-price periods (such as evening peak hours), yielding high economic benefits.

$$P_{cap} = \frac{\sum_{t=1}^{8784} (P_{deliverd,t} \times Price_t \times \Delta t)}{E_{Total}} \tag{14}$$

$P_{delivered,t}$ represents the actual power delivered by the system to the grid or load at time t, calculated as follows:

$$P_{delivered,t} = P_{gen,t} - P_{cur,t} \tag{15}$$

$P_{cur,t}$ represents the curtailment power at time t.

$Price_t$ denotes the electricity market price at time t. According to the 2024 China Power Market Development Report [41], China's average market transaction price is 0.4 yuan per kilowatt-hour. E_{Sold} denotes the system's actual total annual generation.

8 References

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